Clock Regularity and Hybrid Reconstruction for Actual Iterated Brownian Motion

Project Stardust
Private mathematical synthesis

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Abstract

For reflected-clock iterated Brownian motion, source averaging produces multitime covariance functions with folded and collision cusps. We describe their verified membership in the original closed Gaussian clock derivative graph, and the actual predictable Brownian-clock representation of completed Fourier source means. Combining that clock channel with an existing source adjoint reconstructs the entire centered IBM observable through the original intrinsic joint divergence. A sine example shows exactly why reconstructing the mean is insufficient. Hidden-clock predictability, intrinsic observability, and physical-time stochastic calculus remain distinct requirements.

Status. This note summarizes audited Lean results through checkpoint 1520, mathematical commit

`c3381683f4d7abad527dee9575f4cf57deea0431`.

It asserts neither external novelty nor a completed intrinsic IBM calculus. All theorem names below have prefix

`OperatorDerivativeRewrite.Foundation.`

Original research manuscripts remain unchanged. All artifacts are private.

1 The actual process and the averaging operation

Work on the existing independent scenery–clock Gaussian probability space. Write $Z_t = X(|Y_t|)$, where X is the constructed Brownian scenery and Y the constructed Brownian clock. For a finite index set I, nonnegative observation times r_i, real frequencies a_{ki} and phases β_k, let

$$F = \sum_{k\geq 0} w_k \cos\left(\beta_k + \sum_{i\in I} a_{ki} Z_{r_i}\right), \qquad \sum_k |w_k| < \infty.$$

The preexisting Fourier observation is an actual intrinsic IBM L^2 class, identified almost everywhere with this uniformly convergent scalar series. Let $M = \mathbf{E}_X F$ be its preexisting source mean. This is a clock Gaussian L^2 class. Membership in that space alone would not make it an intrinsic function of the observed IBM path.

For $b \in \mathbb{R}^I$ and $x \in \mathbb{R}^I$, the genuine scenery covariance gives

$$Q_b(x) = \sum_{i,j} b_i b_j \min(|x_i|, |x_j|) = \left\| \sum_i b_i h_{|x_i|} \right\|^2 \geq 0,$$

where h_s is the actual scenery first-chaos direction. Consequently

$$M(Y) = \sum_k w_k \cos(\beta_k) \exp(-Q_{a_k}((Y_{r_i})_i)/2) \quad \text{almost everywhere.} \tag{1}$$

Lean reference: `IBMFoldedSourceCovariance gram; variance_expansion`

Lean reference: `IBMFourierClockMeanRegularity clockMean_actual`

This is a law-based source average of the actual observable, not a replacement random variable defined only through a formal operator identity.

2 Cusps in the original closed Gaussian graph

Let D_Y denote the original closed Gaussian derivative and let $e_r = D_Y Y_r$. The existing normalization gives $\|e_r\| = \sqrt{r}$. Products below are almost-everywhere statements about genuine L^2 representatives.

Theorem 2.1 (Absolute value and minimum)**.** *If (f, g) belongs to the original closed graph, then $|f|$ has graph output* $\operatorname{signum}(f)g$*, of norm at most $\|g\|$. If also (h, k) belongs to that graph, then* $\min(f, h)$ *has output*

$$\frac{g + k - \operatorname{signum}(f - h)(g - k)}{2}, \qquad \textit{of norm at most } \|g\| + \|k\|.$$

The convention is $\operatorname{signum}(0) = 0$*; no null-level hypothesis is imposed on either zero sets or collision sets.*

Lean reference: `GaussianAbsoluteMinimumGraph absolute_graph; minimum_graph`

Bounded sine clips of smooth absolute radii converge strongly in both graph coordinates. The proof uses the existing bounded chain rule and closedness of the original operator. It does not introduce a new weak derivative to avoid the domain question.

Put $v_r = \operatorname{signum}(Y_r)e_r$ and

$$H_{rs} = \frac{v_r + v_s - \operatorname{signum}(|Y_r| - |Y_s|)(v_r - v_s)}{2}.$$

Then H_{rs} is the actual graph output of $\min(|Y_r|, |Y_s|)$, and $H_{rr} = v_r$ exactly. The output of $Q_b((Y_{r_i})_i)$ is $V_b = \sum_{i,j} b_i b_j H_{r_i r_j}$. Writing $A(b) = \sum_i |b_i|$, its norm is at most

$$\sum_{i,j} |b_i b_j|(\sqrt{r_i} + \sqrt{r_j}) \le 2A(b)^2 \sum_i \sqrt{r_i}.$$

Signed rows, duplicate clocks, zero times and empty observation sets are included. For $\exp(-Q_b/2)$ the verified graph output is

$$-\tfrac{1}{2} \exp(-|Q_b|/2) \operatorname{signum}(Q_b) V_b.$$

Nonnegativity removes the absolute value in the *value*. The output retains the zero-level sign convention; no second-derivative or divergence domain assertion is inferred from this first graph rule.

Lean reference: `IBMMultitimeClockVarianceGraph overlap_graph; variance_graph; damping_graph`

Theorem 2.2 (Completed source-mean graph)**.** *Assume the value and curvature summability conditions*

$$\sum_k |w_k| < \infty, \qquad \sum_k |w_k| A(a_k)^2 < \infty.$$

Then $M \in \mathrm{Dom}(D_Y)$. Its output is the actual vector series obtained by multiplying each damping output above by $w_k \cos \beta_k$ and summing. Both scalar and vector series are norm summable. The output norm is at most

$$\left(\sum_k |w_k| A(a_k)^2 \right) \sum_i \sqrt{r_i}.$$

Lean reference: `IBMFourierClockGraph`
`multitime_clock_graph; multitime_clock_domain`

The bound is sufficient, not claimed sharp. The stronger single-clock bound proved earlier is preserved. Classical cusp nondifferentiability therefore does not obstruct this original first-order closed graph.

3 A genuine predictable clock identity

Let J_Y be the independently constructed Brownian-clock Itô integral on its actual predictable L^2 space, and let P_Y denote the existing predictable projection of Gaussian vector fields.

Theorem 3.1 (Clock PRP and Clark–Ocone)**.** *For every Fourier observation above with $\sum_k |w_k| < \infty$, the actual mean M lies in the raw Brownian-generated L^2 space. The unique predictable coefficient $U = J_Y^* M$ satisfies*

$$J_Y U = M - \mathbf{E}_Y M. \tag{2}$$

If every $r_i \le T$, U is supported in $[0, T]$, and the original finite-horizon Itô integral gives the same identity. Under the additional curvature summability in Theorem 2.2, $U = P_Y D_Y M$. Coefficients obey $\|U - U'\| \le \|M - M'\|$ in their actual L^2 norms.

Lean reference: `IBMFourierClockClarkOcone`
`stochastic_reconstruction; coefficient_unique`

Lean reference: `IBMFourierClockClarkOcone`
`coefficient_supported; finite_reconstruction`

Lean reference: `IBMFourierClockClarkOcone`
`projected_actual_derivative; coefficient_stability`

The representation, uniqueness and support results do not require curvature summability; that extra hypothesis identifies the coefficient with the full original derivative. Brownian and finite-past measurability are proved from the actual covariance series, not from ambient Gaussian measurability.

4 The actual IBM observable needs both channels

Let $K_X F$ be the preexisting bounded source-adjoint recovery. Let δ_X and δ_Y be the original transported ambient source and clock adjoints. Lift the predictable clock field to the joint space by making it constant in the scenery variable, and call this field $C(U)$. Original adjoint membership and exact source-vacuum identities give

$$\delta_X K_X F = F - M, \qquad \delta_Y C(U) = M - \mathbf{E}_Y M, \qquad \mathbf{E}_Y M = \mathbf{E} F,$$

where clock-only variables are lifted to the product space in these formulas.

Theorem 4.1 (Original intrinsic hybrid reconstruction)**.** *The pair* $(K_X F, C(U))$ *belongs to the domain of the original intrinsic joint divergence* δ_{IBM}*, and*

$$\delta_{\mathrm{IBM}}(K_X F, C(U)) = F - \mathbf{E}F. \tag{3}$$

Its adjoint identity holds against every input in the original intrinsic joint derivative domain. The reconstruction needs only $\sum_k |w_k| < \infty$*.*

Lean reference: `IBMFourierHybridReconstruction`
`source_vacuum; clockMean_expectation`

Lean reference: `IBMFourierHybridReconstruction`
`partial_graph; clock_value; reconstruction; duality`

The last projection onto the actual IBM observable space is legitimate because the centered F is already in that space. There is no assumption that projection preserves derivative domains or predictable fields.

For a concrete test, take one frequency, one nonzero weight and phase $\pi/2$. The actual observable is $F = -\sin(Z_R)$. Its source mean and its entire clock coefficient vanish, but for every $R > 0$ it is nonconstant and outside the original joint derivative domain. Identity (3) still applies through the source channel. Thus (2) cannot be substituted for reconstruction of F itself.

Lean reference: `IBMFourierClockClarkOcone`
`coefficient_zero_of_cos_zero`

Lean reference: `IBMActualSineDomainObstruction`
`sinePoint_not_constant; sinePoint_not_joint_domain`

The precise Fourier/sine identification and the simultaneous positive and negative domain tests are retained in checkpoint 1520's independent Lean checks.

5 What is still missing

The clock coefficient is predictable for the *hidden Brownian clock*. The source channel is the existing static source adjoint, not a claimed physical-time predictable integrand. Neither construction is automatically measurable for the observed IBM filtration. Only U has the uniqueness assertion above; joint divergence integrands can have a kernel.

Other verified results do not remove that distinction. Nonseparable C^1-time/C^4-space Simpson formulas and Bernstein continuous-path realizations hold in their specified probability topologies. The Bernstein construction uses all grid prefixes and is not asserted causal. Separately, the original native grid outputs have no weak intrinsic Hilbert limit, even along a strictly increasing subsequence. Existing Fourier source-lowered actions can converge strongly without contradicting that obstruction: they apply a different tested/lowered map with a specified smaller domain.

Lean reference: `IBMTimeSpaceSimpson`
`sums_tendstoInMeasure`

Lean reference: `IBMContinuousSimpsonRealization`
`approximation_tendstoInMeasure`

Lean reference: `IBMNativeWeakObstruction`
`no_original_weak_subsequence`

Lean reference: `IBMFourierCylinderSeries`
`native_tendsto; source_native_tendsto`

A precise remaining task is to identify an admissible coefficient in the *observed* predictable integral space, prove its actual integral agrees with the desired physical prescription, and show that the centered remainder is zero in the required topology. Alternatively, an explicit nonzero residual would delimit that proposed formula. Hidden-clock representation alone establishes neither conclusion. Nonsmooth physical-time Itô–Tanaka, domain/closure transfers and arbitrary-partition claims retain separate proof obligations. No new campaign, novel-theorem priority or completion claim is created by this synthesis.